MEDRECORDS

Concept Document

A Nationwide Patient-Owned Electronic Medical Record Platform for Nigeria

Document	MedRecords Concept v1.0
Status	Founder Concept Draft
Companion Docs	PRD; Business Model & GTM; Consent & Identity Model; Roadmap; Stress-Test Memo; P

1. Executive Summary

MedRecords is a nationwide, cloud-based Electronic Medical Records (EMR) and Personal Health Record (PHR) platform designed to centralize the medical history of every Nigerian, regardless of where they receive care. Unlike the fragmented, paper-based system currently used across Nigeria, MedRecords creates a single, patient-owned, portable medical identity that follows the individual across hospitals, clinics, pharmacies, dentists, eye-care centers, labs, and other healthcare providers.

The platform is built as a multi-channel SaaS solution accessible via web applications, mobile apps (iOS / Android), desktop applications, and API integrations for hospitals and pharmacies.

MedRecords enables secure, consent-driven access to a patient's medical history while enforcing role-based visibility so that each provider only sees what they are authorized to see.

2. Core Problem in Nigeria's Healthcare System

Current State

- Medical records are handwritten and stored physically in individual hospitals.
- Patients have no centralized or portable medical history.
- Records are frequently lost, incomplete, or inaccessible.
- Pharmacies, labs, dentists, and specialists operate in isolation.
- Embassies and international institutions often require medical history that Nigerians cannot easily provide.

Impact

- Doctors cannot make informed decisions due to lack of history.
- Patients repeat tests unnecessarily.
- Medication errors occur because pharmacies cannot see prescriptions.
- No continuity of care across providers.
- No digital trail for surgeries, diagnoses, allergies, or chronic conditions.

3. Vision & Value Proposition

Vision

To create Nigeria's first unified, secure, and patient-controlled medical record system, enabling seamless healthcare delivery across all providers.

Value Proposition

For Patients

A lifetime medical history accessible anytime, anywhere — including prescriptions, surgeries, diagnoses, allergies, lab results, dental records, eye prescriptions, and more.

For Doctors & Hospitals

Instant access to accurate patient history (with consent), improving diagnosis, reducing errors, and enabling continuity of care.

For Pharmacies

Ability to verify prescriptions, record dispensed medications, and maintain compliance.

For Embassies & International Travel

Exportable, verifiable medical history for visa applications, immigration, and international treatment.

4. Platform Features

A. Patient-Centered Medical Identity

Each Nigerian has a unique MedRecords ID that links all their medical interactions across hospitals, clinics, pharmacies, dentists, eye-care centers, chiropractors, laboratories, and emergency services. The patient owns the account and controls access.

B. Consent-Based Access Control

Before any provider can view a patient's history, the patient must grant consent digitally or via a signed form. Consent types: full access (general practitioners, primary hospitals); partial access (specialists, dentists, pharmacies); time-limited access (e.g., 24 hours for emergency care).

C. Role-Based Data Visibility

Different providers see only what is relevant to their specialty:

- Dentist: only dental history, X-rays, procedures, and medications related to oral care.
- Pharmacy: only prescriptions, medication history, and allergies.
- Eye-care doctor: eye exams, prescriptions, diagnoses.
- General practitioner: full medical history (if consent is granted).
- Chiropractor: musculoskeletal diagnoses, therapy notes.

D. Medical Record Categories

Each patient profile contains structured tabs:

- General Medical History
- Surgeries & Procedures

- Hospitalizations
- Prescriptions & Medications
- Pharmacy Dispensing Records
- Dental Records
- Eye Care & Optical Prescriptions
- Lab Results & Diagnostics
- Vaccination History
- Allergies & Chronic Conditions
- Emergency Notes
- Referrals & Specialist Reports

5. Technical Architecture (High-Level)

Frontend

React (web), React Native / Flutter (mobile), Electron (desktop).

Backend

Node.js / .NET Core microservices. REST + GraphQL APIs. Event-driven architecture (Kafka or RabbitMQ).

Database Layer

- PostgreSQL for structured medical data.
- MongoDB for unstructured notes.
- S3 / Blob storage for images, scans, X-rays.

Identity & Access Management

OAuth 2.0 / OpenID Connect. Multi-factor authentication. Role-based access control (RBAC). Attribute-based access control (ABAC) for specialty restrictions.

Security

- End-to-end encryption.
- AES-256 at rest.
- TLS 1.3 in transit.
- Audit logs for every access.
- Consent tokens with expiration.

Integration Layer

APIs for hospitals, pharmacies, labs. SDKs for third-party developers. HL7 / FHIR compatibility for global interoperability.

6. Data Flow Example — Patient Visits a Dentist

1. Patient provides MedRecords ID.
2. Dentist requests access; patient approves.
3. Dentist sees only the dental tab.
4. Dentist uploads X-rays, diagnosis, and procedure notes.
5. Data is stored in the patient's central record.
6. Patient can view it anytime.
7. A general doctor (with full consent) can see this dental record later.

7. Use Cases

A. Patient

- View lifetime medical history.
- Download records for embassy or visa applications.
- Share records with new doctors.
- Track medications and allergies.
- Manage dependents' records (children, elderly parents).

B. Hospital

- Access patient history before treatment.
- Record surgeries, diagnoses, admissions.
- Reduce repeated tests.
- Improve emergency response.

C. Pharmacy

- Verify prescriptions.
- Record dispensed medications.
- Prevent drug interactions.

D. Specialists

- Dentists, eye doctors, chiropractors, and others access only relevant medical categories.
- Upload specialty-specific records.

8. Compliance & Governance

MedRecords will align with: the Nigeria Data Protection Regulation (NDPR) and successor Nigeria Data Protection Act 2023 (NDPA); HIPAA-inspired security standards; international FHIR / HL7 interoperability frameworks; and digital consent and audit trail requirements.

9. Future Expansion

- AI-driven diagnosis support.
- Drug interaction alerts.
- Telemedicine integration.
- Insurance claims integration.
- National health ID linkage.
- Offline mode for rural clinics.

10. Summary

MedRecords is a transformative platform that solves one of Nigeria's biggest healthcare challenges: the absence of centralized, secure, and portable medical records. By giving patients ownership of their medical history and enabling providers to access accurate information with consent, MedRecords modernizes the entire healthcare ecosystem.